LUCAS MARQUES SNELLER

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EDUCATION

Columbia University New York, NY
Masters of Science in Quantitative Methods in the Social Sciences, (Economics Focus) Expected May 2027

Miami University Oxford, OH
Bachelor of Science in Political Science and Philosophy August 2022-May 2026

- Dual Major with an Economics Minor
- Awards: HASS Scholar, Prodesse Scholar

RELEVANT EXPERIENCE

HASS Undergraduate Research Program Oxford,OH
Undergraduate Researcher 2022-Present

- Utilize transferable skills and conduct extensive undergraduate research and presented research at symposium regarding DEI programs and availability in predominantly white institutions

Miami University Oxford,OH
Research Assistant 2025-Present

- Collaborate with faculty across Economics, Political Science, and Computer Science to support multiple concurrent research initiatives, translating research questions into reproducible data/analysis workflows, maintaining clear documentation, and delivering polished research outputs (projects detailed below).

Owens Corning Financial Case Study Competition Winner Oxford, OH
Spring 2023

- Awarded first place for study of international markets and provided the best new revenue sources

PROJECTS

- **LLM SOC Classification (Research Assistant):** Built an LLM-driven SOC-2018 classifier using Retrieval Augmented Generation (RAG), hybrid dense+sparse retrieval, Prediction Powered Inference (PPI), and exemplar prompting; designed a calibrated decision layer and prediction-powered inference with audit tooling.
- **Asian American History Project (Research Assistant):** Supported an initiative to document and digitize Asian and Asian American history in Ohio; ran OCR on scanned archives, built embeddings/vector indexes, and prototyped a multilingual RAG+LLM Q&A platform for researchers and community partners.
- **Economic Impact of Butler County Parks (Research Assistant):** Collaborated with faculty and local partners on a parcel-level GIS and hedonic pricing model; linked park proximity to housing sale prices, property tax uplift, and a tract-level Park Access Index tied to health outcomes, civic indicators, and quantitative estimates of parks' contributions to community well-being.
- **Double Machine Learning for Dynamic Panels:** Developed a DML inference framework for high-dimensional time series using Neyman-orthogonal partialling-out scores, blocked-time cross-fitting with buffering, and unit-clustered variance estimation; established asymptotic normality under weak dependence and validated improved bias reduction through Monte Carlo stress-testing
- **Intermediary Asset Pricing & Sentiment:** Developed an endogenous two-state Markov Regime Switching model to identify intermediary constraints from aggregate returns; demonstrated that negative sentiment shocks in constrained regimes trigger liquidity-driven crashes and predictable reversals in high-inventory-risk stocks, quantifying a $384 billion wealth transfer from forced sellers.
- **Mean Field Game Dynamics:** Formulated a dynamic MFG framework coupling endogenous price formation with overconfident Bayesian filtering; derived a closed-form linear feedback policy under Gaussian increments and utilized particle-based simulations to demonstrate that miscalibrated beliefs amplify trading intensity and cross-sectional disagreement.

SKILLS

- **Analytical Skills** - Applied econometric and statistical modeling (GIS hedonic pricing, dynamic-panel DML, Markov regime switching, MFG simulation) to analyze complex datasets, isolate root drivers, and design validated, performance-improving solutions supported by robustness and Monte Carlo testing.
- **Writing and Verbal Communication** - Authored technical reports, model documentation, and executive-ready slide decks; delivered research presentations at symposiums and competitions; and facilitated cross-functional meetings to translate quantitative findings into clear narratives and next-step decisions.
- **Technical Expertise** - Proficient in Excel/Word/PowerPoint and quantitative stacks (Python/R; MATLAB/STATA familiarity); built reproducible data pipelines, vector-search/RAG prototypes, and econometric simulation tooling, with strong versioning practices and rapid onboarding to new technologies.
- **Language** - Classroom Spanish and family exposure to Brazilian Portuguese; supported multilingual archival OCR/embedding workflows for research and community partners.
- **Interests** - Economics and econometrics, applied mathematics and dynamical systems, stochastic control/mean field games, AI/LLM development for research automation, mathematical modeling, statistical inference, and decision theory.